

## CS23334-FUNDAMENTALS OF DATA SCIENCE

### UNIT 4 ASSIGNMENT

**Instructions:** Answer all questions. For each problem, state  $H_0$  and  $H_1$ , the level of significance, the test statistic, the table value, the decision and the conclusion.

#### Z-Test

1. For the population at large, the Wechsler Adult Intelligence Scale is designed to yield a normal distribution of test scores with a mean of 100 and a standard deviation of 15. School district officials wonder whether, on the average, an IQ score different from 100 describes the intellectual aptitudes of all students in their district. Wechsler IQ scores are obtained for a random sample of 25 of their students, and the mean IQ is found to equal 105. Using the step-by-step procedure, test the null hypothesis at the .05 level of significance. **(Two-tailed)**

**Table value:**  $Z$  (two-tailed,  $\alpha = 0.05$ ) =  $\pm 1.96$

2. A tea stall in Chennai sold an average of 400 cups per day with a standard deviation of 45 cups. After introducing a new flavour, the owner records sales for a random sample of 36 days and finds a mean of 418 cups per day. Test at the 0.05 level of significance whether the new flavour has **increased** daily sales. **(Right-tailed)**

**Table value:**  $Z$  (right-tailed,  $\alpha = 0.05$ ) =  $+1.645$

3. A battery manufacturer claims that the mean life of its batteries is at least 300 hours, with a known standard deviation of 30 hours. A consumer forum tests a random sample of 40 batteries and finds a mean life of 291 hours. At the 0.05 level of significance, is there evidence that the mean life is **less** than claimed? **(Left-tailed)**

**Table value:**  $Z$  (left-tailed,  $\alpha = 0.05$ ) =  $-1.645$

4. In a university examination, a random sample of 100 students from College A had a mean score of 68 (population SD = 10), and a random sample of 120 students from College B had a mean score of 65 (population SD = 12). Test at the 0.05 level of significance whether there is a significant difference between the mean scores of the two colleges. **(Two-sample Z-test)**

**Table value:**  $Z$  (two-tailed,  $\alpha = 0.05$ ) =  $\pm 1.96$

#### t-Test: One Sample

5. A library system lends books for periods of 21 days. This policy is being reevaluated in view of a possible new loan period that could be either longer or shorter than 21 days. To aid in making this decision, book-lending records were consulted to determine the loan periods actually used by the patrons. A random sample of eight records revealed the following loan periods in days: 21, 15, 12, 24, 20, 21, 13, and 16. Test the null hypothesis with t-test, using the .05 level of significance.

**Table value:**  $t$  (two-tailed,  $\alpha = 0.05$ ,  $df = 7$ ) =  $\pm 2.365$

6. The average growth of a certain variety of pine trees is 10.1 inches in three years. A biologist claims that a new variety will have improved three-year growth. A random sample of 25 of the new variety has an average three-year growth of 10.8 inches and a standard deviation of 2.1 inches. Test this against the biologist's claim with 0.1 level of significance.

**Table value:**  $t$  (one-tailed,  $\alpha = 0.10$ ,  $df = 24$ ) = 1.318

7. A soft-drink company fills bottles labelled 500 ml. The quality inspector measures 10 randomly selected bottles and records the following volumes (ml): 498, 502, 495, 497, 499, 494, 500, 496, 498, 495. Test at the 0.05 level of significance whether the mean volume differs from 500 ml.

**Table value:**  $t$  (two-tailed,  $\alpha = 0.05$ ,  $df = 9$ ) =  $\pm 2.262$

### t-Test: Two Independent Samples

8. An investigator wishes to determine whether alcohol consumption causes a deterioration in the performance of automobile drivers. Before the driving test, subjects drink a glass of orange juice, which, in the case of the treatment group, is laced with two ounces of vodka. Performance is measured by the number of errors made on a driving simulator. A total of 120 volunteer subjects are randomly assigned, in equal numbers, to the two groups. For subjects in the treatment group, the mean number of errors  $\bar{X}_1$  equals 26.4, and for subjects in the control group, the mean number of errors  $\bar{X}_2$  equals 18.6. The estimated standard error equals 2.4.

(i) Use  $t$  to test the null hypothesis at the .05 level of significance.

(ii) Specify the  $p$ -value for this test result.

(iii) If appropriate, construct a 95 percent confidence interval for the true population mean difference and interpret this interval.

(iv) If the test result is statistically significant, use Cohen's  $d$  to estimate the effect size, given that the standard deviation,  $s_p$ , equals 13.15.

**Table value:**  $t$  (two-tailed,  $\alpha = 0.05$ ,  $df = 118$ ) =  $\pm 1.980$  (also used for the 95% confidence interval)

9. Two sections of a class were taught statistics by different methods. The marks obtained by randomly selected students are given below. Test at the 0.05 level of significance whether there is a significant difference between the mean marks of the two sections.

|                           |    |    |    |    |    |    |    |    |    |    |
|---------------------------|----|----|----|----|----|----|----|----|----|----|
| <b>Section A (n = 8)</b>  | 72 | 68 | 75 | 80 | 70 | 74 | 69 | 77 |    |    |
| <b>Section B (n = 10)</b> | 65 | 70 | 62 | 68 | 72 | 66 | 64 | 69 | 71 | 63 |

**Table value:**  $t$  (two-tailed,  $\alpha = 0.05$ ,  $df = 16$ ) =  $\pm 2.120$

### t-Test: Two Related (Paired) Samples

10. A manufacturer of a gas additive claims that it improves gas mileage. A random sample of 30 drivers tests this claim by determining their gas mileage for a full tank of gas that

contains the additive ( $X_1$ ) and for a full tank of gas that does not contain the additive ( $X_2$ ). The sample mean difference,  $\bar{D}$ , equals 2.12 miles (in favor of the additive), and the estimated standard error equals 1.50 miles.

- (i) Using t, test the null hypothesis at the .05 level of significance.
- (ii) Specify the p-value for this result.
- (iii) Are there any special precautions that should be taken with the present experimental design?

**Table value:**  $t$  (one-tailed,  $\alpha = 0.05$ ,  $df = 29$ ) = 1.699

11. The typing speed (words per minute) of 8 office employees was recorded before and after a two-week training programme. Test at the 0.05 level of significance whether the training has **increased** typing speed.

| Employee | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|----------|----|----|----|----|----|----|----|----|
| Before   | 42 | 38 | 45 | 50 | 36 | 40 | 48 | 44 |
| After    | 48 | 41 | 47 | 55 | 40 | 46 | 50 | 49 |

**Table value:**  $t$  (one-tailed,  $\alpha = 0.05$ ,  $df = 7$ ) = 1.895

### F-Test for Equality of Variances

12. Two machines produce the same component. A random sample of 11 components from Machine A gives a standard deviation of 2.4 mm, and a random sample of 9 components from Machine B gives a standard deviation of 1.5 mm. Test at the 0.05 level of significance whether the two machines have the same variability. **(Two-tailed)**

**Table value:**  $F$  ( $\alpha/2 = 0.025$ ,  $df_1 = 10$ ,  $df_2 = 8$ ) = 4.30

13. The life (in hours) of batteries of two brands is given below. Test at the 0.05 level of significance whether the variances of the two brands are equal.

|                        |    |    |    |    |    |    |    |    |
|------------------------|----|----|----|----|----|----|----|----|
| <b>Brand X (n = 8)</b> | 42 | 45 | 39 | 48 | 41 | 50 | 38 | 47 |
| <b>Brand Y (n = 7)</b> | 44 | 45 | 43 | 46 | 44 | 45 | 43 |    |

**Table value:**  $F$  ( $\alpha/2 = 0.025$ ,  $df_1 = 7$ ,  $df_2 = 6$ ) = 5.70

### ANOVA: One-way

14. A catering firm wishes to buy a meat tenderizer, but was concerned with the effect on the weight loss of meat during cooking. The following results were obtained for the weight loss of steaks of the same pre-cooked weight when three different tenderizers were used.

| Tenderiser | Weight loss in grams |
|------------|----------------------|
| A          | 36 28 42 58          |
| B          | 17 59 33             |
| C          | 36 74 29 55 48       |

Carry out one factor analysis of variance and test at the 5% significance level whether there is a difference in weight loss between tenderisers.

**Table value:**  $F(\alpha = 0.05, df_1 = 2, df_2 = 9) = 4.26$

15. Three teaching methods were tried on three groups of 5 students each. Their test scores are given below. Construct the ANOVA table and test at the 0.05 level of significance whether the three methods give the same mean score.

|            |    |    |    |    |    |
|------------|----|----|----|----|----|
| Method I   | 70 | 75 | 72 | 68 | 74 |
| Method II  | 80 | 78 | 85 | 82 | 79 |
| Method III | 65 | 70 | 68 | 72 | 66 |

**Table value:**  $F(\alpha = 0.05, df_1 = 2, df_2 = 12) = 3.89$

### ANOVA: Two-way

16. Four varieties of wheat (V1 to V4) were grown with three different fertilizers (F1 to F3). The yields (kg per plot) are given below, with one plot for each combination. Carry out a two-way ANOVA and test at the 0.05 level of significance whether there are significant differences (i) between the varieties and (ii) between the fertilizers. **(Without replication)**

| Variety | F1 | F2 | F3 |
|---------|----|----|----|
| V1      | 20 | 22 | 24 |
| V2      | 18 | 21 | 23 |
| V3      | 25 | 24 | 27 |
| V4      | 22 | 23 | 26 |

**Table value:** Varieties:  $F(\alpha = 0.05, df_1 = 3, df_2 = 6) = 4.76$ ; Fertilizers:  $F(\alpha = 0.05, df_1 = 2, df_2 = 6) = 5.14$

17. A company advertised its product through three media (TV, Radio, Newspaper) in two cities (Chennai, Coimbatore). The weekly sales (units) for three weeks in each combination are given below. Carry out a two-way ANOVA at the 0.05 level of significance to test (i) the effect of advertising media, (ii) the effect of city and (iii) the interaction between media and city. **(With replication)**

| Media     | Chennai    | Coimbatore |
|-----------|------------|------------|
| TV        | 45, 48, 50 | 40, 42, 43 |
| Radio     | 38, 36, 40 | 35, 37, 34 |
| Newspaper | 30, 33, 32 | 31, 29, 30 |

**Table value:** Media:  $F(\alpha = 0.05, df_1 = 2, df_2 = 12) = 3.89$ ; City:  $F(\alpha = 0.05, df_1 = 1, df_2 = 12) = 4.75$ ; Interaction:  $F(\alpha = 0.05, df_1 = 2, df_2 = 12) = 3.89$

### Chi-square Tests

18. A random sample of 90 college students indicates whether they most desire love, wealth, power, health, fame, or family happiness. Using the .05 level of significance and the following results, test the null hypothesis that, in the underlying population, the various desires are equally popular using chi-square test. **(Goodness of fit)**

| Desire             | Love | Wealth | Power | Health | Fame | Family Hap. | Total |
|--------------------|------|--------|-------|--------|------|-------------|-------|
| Observed ( $f_o$ ) | 25   | 10     | 5     | 25     | 10   | 15          | 90    |

**Table value:**  $\chi^2(\alpha = 0.05, df = 5) = 11.070$

19. A psychologist is observing eating behaviour in 131 children aged 3 years old from Chennai. He presents each child 20 new foods which they have never eaten before. He then records the number of foods they actually try. The results are shown in the table below.

| Number of new foods tried | Frequency |
|---------------------------|-----------|
| 0 – 5                     | 45        |
| 6 – 10                    | 34        |
| 11 – 15                   | 31        |
| 16 – 20                   | 21        |

Previous research with children from across the state has shown that we expect 40% of young children to try 0 to 5 new foods, 30% to try 6 to 10 new foods, 20% to try 11 to 15 new foods and 10% to try 16 to 20 new foods. Perform a chi-square test with 0.05 significance level to see if the children from Chennai follow the same distribution that the research on State children has found. **(Goodness of fit)**

**Table value:**  $\chi^2(\alpha = 0.05, df = 3) = 7.815$

20. The placement records of 220 final-year students from three departments are given below. Test at the 0.05 level of significance whether placement status is independent of the department. **(Test of independence)**

| Status     | CSE | ECE | MECH |
|------------|-----|-----|------|
| Placed     | 60  | 45  | 30   |
| Not placed | 20  | 25  | 40   |

**Table value:**  $\chi^2 (\alpha = 0.05, df = 2) = 5.991$

21. In a study of 400 people, the following data were collected on vaccination and infection. Test at the 0.05 level of significance whether infection is associated with vaccination status. **(Test of independence,  $2 \times 2$  table)**

|                | Infected | Not infected | Total |
|----------------|----------|--------------|-------|
| Vaccinated     | 15       | 185          | 200   |
| Not vaccinated | 45       | 155          | 200   |

**Table value:**  $\chi^2 (\alpha = 0.05, df = 1) = 3.841$